Group 21

Toronto Bikeshare Demand Forecasting

Abdul Mohammed[1] Axel Tang[2] Esam Uddin[3]

[1]The Edward S. Rogers Sr. Department of Electrical and Computer Engineering

Introduction to Machine Learning – Winter 2026

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Slide 1

Project Overview

- Goal: forecast Toronto Bike Share demand at hourly resolution.
- Data scope: 2025 to early 2026 ridership, station metadata, and hourly weather.
- Core workflow: data cleaning $\rightarrow$ EDA $\rightarrow$ feature engineering $\rightarrow$ baseline model $\rightarrow$ tree-based models.
- Evaluation focus: MAE, RMSE, and R^2 on held-out test period.

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Slide 2

Data Cleaning and Ingestion

- Merged monthly ridership files for 2025–2026.
- Dropped rows with missing start station/time.
- Filtered trip duration to 60−−86400 seconds.
- Aggregated to station-hour pickups and filled implicit zeros.
- Cleaned station metadata: **1031 stations**.
- Weather cleanup produced **0 nulls** in key columns.

Output of This Stage

Prepared consistent modeling files in `data/processed/` (hourly demand, clusters, weather, train/val/test).

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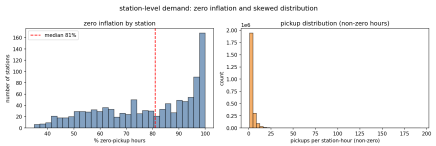
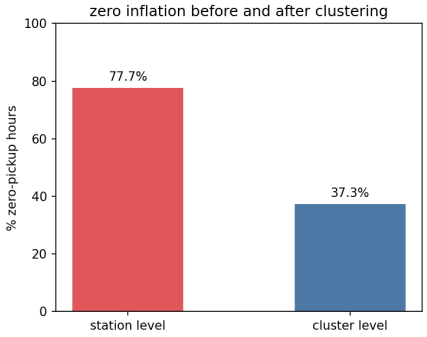
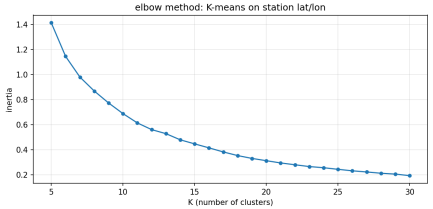
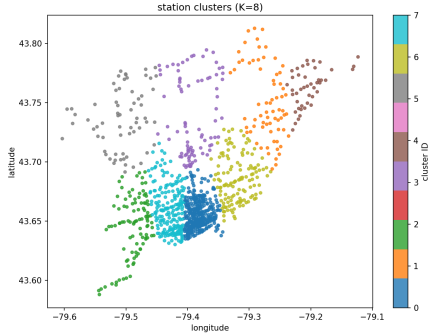
EDA Results

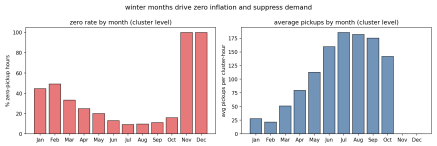
- EDA showed strong zero inflation and clear seasonal/hourly structure.
- Used K-means clustering with $K = 8$ for station grouping.
- Diagnostics shown below: zeros, cluster geometry, seasonal demand, and hour-month patterns.

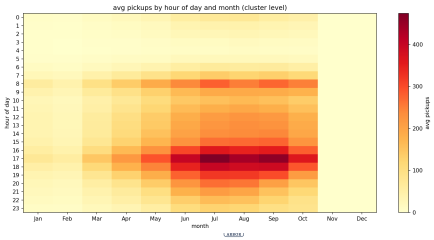








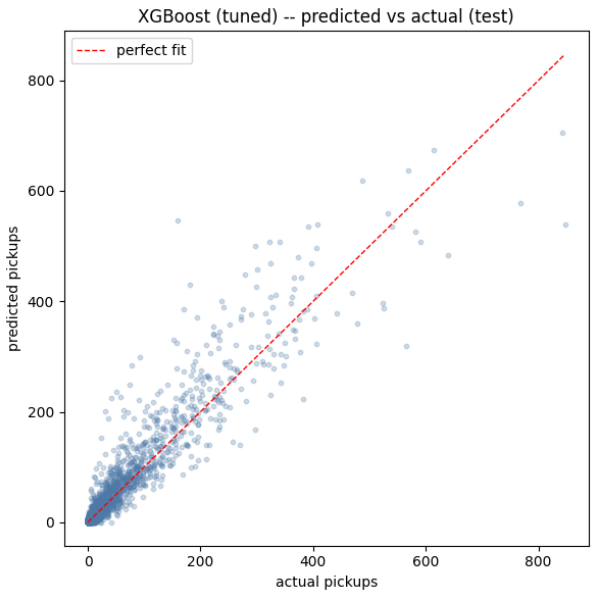




Random Forest and XGBoost Results

Model	MAE	R^2	RMSE
Linear Regression	11.1769	0.7083	33.0073
Random Forest (baseline)	7.5952	0.8504	23.6354
Random Forest (tuned)	7.4770	0.8529	23.4433
XGBoost (baseline)	7.3418	0.8647	22.4805
XGBoost (tuned)	**7.1478**	**0.8701**	**22.0275**

- Best validation model: **XGBoost (tuned)**.
- Best test model: **XGBoost (tuned)**.
- Clear improvement over linear baseline on all three metrics.



XGBoost (tuned) predicted vs actual (test).